MATH 1271 **Name:** _____
Fall 2026
Quiz 13: 10/17 Content
10/29/2026
Time Limit: 5 Minutes **Student ID:** _____

This quiz consists of 1 questions across 2 pages worth 5 points.
Rules:

- You have the indicated amount of time above to complete the quiz and you must not begin working on the quiz until instructed to do so.
- You must show all of your work to receive full credit, unless stated otherwise. **On problems that require work to be shown, answers that are mysterious or unsupported will not receive full credit.**
- You may not use a calculator, textbook, notes or any other external resources.
- Please write your initials at the top of every page in the event the pages become separated.
- If you run out of space on a page, feel free to use the back of the page to continue your work.
- You understand that if you violate any of these or the University's rules, you will receive a score of zero on this quiz and you may be subject to further disciplinary action.

Following the University of Minnesota's policy on academic honesty, I pledge that the work shown on this quiz is entirely my own and that I have followed the rules stated above.

Signature: _____

Question	Points
1	5
Total:	5

1. (5 points) Compute the limit

$$\lim_{x\to\infty} \frac{\ln^2(x)}{x} \tag{1}$$

using L'Hôpital's rule.

Solution: We first need to determine if the limit is in a form where L'Hôpital's rule applies. As $x \to \infty$, we know that $\ln(x) \to \infty$ too, thus this limit is an indeterminate form of type $\frac{\infty}{\infty}$, and thus we can apply L'Hôpital's rule.

Since $\frac{d}{dx}\ln^2(x) = \left(\frac{d}{du}\,|_{u=\ln(x)}\; u^2\right) \cdot \left(\frac{d}{dx}\ln(x)\right) = 2\ln(x)\cdot\frac{1}{x}$ and $\frac{d}{dx}x = 1$ we have that

$$\lim_{x\to\infty} \frac{\ln^2(x)}{x} = \lim_{x\to\infty} \frac{\frac{d}{dx}\ln^2(x)}{\frac{d}{dx}x} \tag{2}$$

$$= \lim_{x\to\infty} \frac{2\ln(x)\cdot\frac{1}{x}}{1}. \tag{3}$$

When we try and analyze this limit, we note that it's current form is hard to interpret and so we simplify the fraction to get

$$\lim_{x\to\infty} \frac{2\ln(x)\cdot\frac{1}{x}}{1} = \lim_{x\to\infty} \frac{2\ln(x)}{x} \tag{4}$$

which, by the above reasoning, is indeterminate of type $\frac{\infty}{\infty}$.

We again apply L'Hôpital's rule to get:

$$\lim_{x\to\infty} \frac{2\ln(x)}{x} = \lim_{x\to\infty} \frac{\frac{d}{dx}2\ln(x)}{\frac{d}{dx}x} \tag{5}$$

$$= \lim_{x\to\infty} \frac{2\cdot\frac{1}{x}}{1} \tag{6}$$

$$= \lim_{x\to\infty} \frac{2}{x} = 0. \tag{7}$$

Therefore, we have that:

$$\lim_{x\to\infty} \frac{\ln^2(x)}{x} = 0. \tag{8}$$